



Distribution of inhaled volatile turmerones in the mouse

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ABSTRACT

Turmerones (α -turmerone, β -turmerone, and ar-turmerone), the main bioactive compounds in turmeric (*Curcuma longa*), are used as herbal medicines in Asia. The distinct feature of turmerones is their strong flavor, which is attributed to their volatility. Orally administered turmerones are reportedly distributed in their intact forms in organs; however, the distribution of inhaled volatile turmerones in the body remains unknown. This study aimed to estimate the distribution of inhaled volatile turmerones. After mice were exposed to volatile turmeric oil, turmerones were found in the lung, olfactory bulb, brain, serum, heart, liver, kidney, epididymal fat, and brown adipose tissue. Each turmerone had a different range of concentrations in the different organs: α -turmerone (5.74–425 ng/kg), β -turmerone (6.41–231 ng/kg), and ar-turmerone (61.0–594 ng/kg). These results suggested that volatile turmerones could be incorporated into the animal body with a distinct range of concentration for each organ.

1. Introduction

Turmeric (*Curcuma longa*) is a perennial herb of the ginger family that mainly originated from India. It is used as a coloring agent, spice, or flavoring agent. In traditional Chinese and Ayurvedic medicines, the turmeric rhizome has been used as a herbal medicine to promote liver function and digestion (Chaturvedi, 2009; Gupta et al., 2013). Turmerones (α -turmerone, β -turmerone, and ar-turmerone) are one of characteristic compounds in turmeric. (Xian et al., 1998). *In vitro* studies have suggested that turmerones have several biological activities, such as anti-inflammatory (Li et al., 2018; Park et al., 2012a, 2012b), anti-immunomodulatory (Yue et al., 2010), anticancer (Mingjie et al., 2004; Shao et al., 2012) and antifungal activities (Onkar et al., 2007), and inhibitory effects on β -secretase which is a potential contributor to the development of Alzheimer's disease (Matsumura et al., 2016). *In vivo* studies have suggested that turmerones have several biological activities, such as an antidepressant (Jung et al., 2013), a therapeutic drug for epilepsy (Adriana et al., 2012), a blood glucose lowering effect (Kuroda et al., 2005), antioxidant activity (Vijayastelter et al., 2011), and induction of neuronal differentiation of neural stem cells (Hucklenbroich et al., 2014).

In addition to the biological activities, the volatility of turmerones distinguishes them from the functional compounds of turmeric. Turmerones are volatilized and have a strong flavor. Therefore, people also inhale volatile turmerones as much as people eat them. People who work in food factories often inhale volatile turmerones, when extracting active ingredients from turmeric. Oral administered turmerones are reportedly distributed in their intact forms in the organs (Iwamoto et al., 2019). However, there have been no studies on the distribution properties of inhaled volatile turmerones in the body. In this study, the distribution properties of inhaled volatile turmerones were estimated.

2. Materials and methods

2.1. Materials

The rhizomes of *C. longa* Linne were purchased from Arjuna Natural Pvt. Ltd. (Alwaye, India). Fresh rhizomes of *C. longa* Linne, the cultivated variety IISR Alleppey Supreme, were collected from a farm in Alleppey, Kerala, India and they were dried according to Jayashree et al. (2015). The rhizomes were authenticated by a qualified botanist at the Arjuna Natural Pvt. Ltd. (ID –AE-HBRS-22). Cleaned rhizomes were fed into the Mini pulveriser, a back loading type (PILOT 612; Pilotsmith

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List of abbreviations

CIS	cooled injection system
GC-MS	gas chromatograph-mass spectrometer
I.D.	inner diameter
LC-MS/MS	liquid chromatography-tandem mass spectrometry
N.D.	not detected
ODS	octadecylsilyl
TDS	thermal desorption system

India Pvt. Ltd., Kerala, India). The disintegrated material from the outlet was collected and sieved through a 14–16 mesh screen. The crude turmeric oil (310 g) was obtained from the milled rhizome (10 kg) using steam distillation in a Clevenger-type apparatus (Magnum Glass Works, Kerala, India). The crude turmeric oil (100 g) was distilled under low pressure (boiling point 120 °C at 150 Pa) to remove compounds with low-boiling points. The oil was further distilled at >150 °C to obtain a turmerone-rich fraction (74 g) of turmeric oil. Turmeric oil was stored at 4 °C in brown bottle for a maximum of 6 wk. Turmerones used as authentic samples for quantitation were purified from rhizomes using a n-hexane extract as follows: The extract was subjected to preparative HPLC (LC-8A system; Shimadzu Co., Ltd., Kyoto, Japan) with these conditions: column (L-Column octadecylsilyl (ODS); length: 250 mm, particle size: 5 µm, inner diameter (I.D.): 20 mm, Chemicals Evaluation and Research Institute, Tokyo, Japan); mobile phase: water:acetonitrile (20:80, v/v); flow rate: 18.9 mL/min; detection: UV 190 and 243 nm; retention time, 12.5 min for α -turmerone, 11.4 min for β -turmerone and 8.4 min for ar-turmerone.

2.2. GC-MS analysis

The turmerones concentration were measured using a gas chromatograph-mass spectrometer (GC-MS) (Agilent 6890A-5973N; Agilent Technologies, Santa Clara, CA, USA). Turmeric oil (10 mL) was soaked in absorbent cotton and placed in a net (mesh: 0.18 cm², Hatto Co., Wakayama, Japan). It was hung using a paper string (Kouyu Co., Ltd., Fukushima, Japan) and the flask was closed with a silicone stopper. After 0, 10, and 60 min, turmerones were adsorbed using a Tenax TA cartridge (length: 17.8 cm, outer diameter: 6 mm, I.D.: 4 mm, Sigma Aldrich Co., St. Louis, MO, USA) at 3 cm from the bottom of the flask to sample 100 mL of air. The absorbed turmerones in the Tenax TA cartridge were extracted in a thermal desorption system (TDS) (TDS 2; Gerstel K.K., Mülheim an der Ruhr, Germany). The parameters of the TDS were as follows: transfer temperature: 260 °C, initial temperature: 20 °C, initial time: 0 min, rate: 60 °C/min, final temperature: 240 °C, final time: 5 min. Next, turmerones were transferred to a cooled injection system (CIS) (CIS 4; Gerstel K.K.) to trap volatile turmerones before GC-MS analysis. The parameters of the CIS were as follows: initial temperature: -100 °C, initial time: 0.05 min, rate: 12.0 °C/s, final temperature: 240 °C, final time: 1 min. For GC-MS analysis, the InertCap Pure WAX column (length: 60 m, outer diameter: 0.2505 mm, I.D.: 0.25 mm, GL Sciences, Inc., Tokyo, Japan) was used for the separation. The oven temperature was programmed as follows: initial temperature: 180 °C, rate: 5 °C/min (180–220 °C), 10 °C/min (220–240 °C), final temperature: 240 °C for 8.5 min. The He inlet pressure was controlled with an electronic pressure control system to achieve a constant column flow of 1.0 mL/min. Mass spectrometry (MS) analysis was done in the electron ionization (EI) mode at a voltage of 70 eV. Using the peak area of each turmerone, the turmerones concentration/100 mL was determined from each of their standard curves. Quantification of α -turmerone, β -turmerone and ar-turmerone was done using the strongest fragment peak, i.e., m/z 83 for α -turmerone, m/z 120 for β -turmerone, and m/z 83 for ar-turmerone. Retention time was 8.5 min for

α -turmerone, 9.1 min for β -turmerone, and 9.2 min for ar-turmerone.

2.3. Animals

All the animal experiments were approved by the Institutional Animal Care and Use Committee and done according to the Kindai University Animal Experimentation Regulations (approval number KAAG-31-008). All efforts were made to minimize suffering. Five wk old male ddY mice (Japan SLC, Inc., Shizuoka, Japan) were provided with free access to a commercial diet (MF; water: 7.9%, crude protein: 23.1%, crude fat: 5.1%, crude ash: 5.8%, crude fiber: 2.8%, and solubles nitrogen-free: 55.3%, Oriental Yeast Co., Ltd., Tokyo, Japan) and water. They were maintained at 25 ± 1 °C on a standard 12 h light/dark cycle.

2.4. Evaluation of the distribution of turmerones in mice

After 5 days of habituation, the mice were divided into different groups. The negative control group (no exposure group (n = 7)), 60–0 min group (volatile turmeric oil exposure for 60 min (first number) and 0 min breathing (second number after the hyphen) in normal air (n = 6)), 60–60 min group (n = 6), 60–180 min group (n = 6), 60–24 h group (n = 6), and 60–72 h group (n = 7). The experimental scheme is shown in [Supplementary Figure S1\(a\)](#). The mice in each group were placed in a flask for 60 min with or without volatile turmerones ([Supplementary Figure S1\(b\)](#)). In preparation for the turmeric oil-exposed groups, samples were prepared as in section 2.2 and a 5000 mL flask was closed with a silicone stopper for 10 min to fill it with volatile turmerones. After 10 min, the mice were anesthetized using sodium pentobarbital (50 mg/kg (bw), intraperitoneal injection, Nacalai Tesque, Inc., Kyoto, Japan), and put into the flask with the silicone stopper for 60 min to expose the mice to volatile turmerones. After 60 min of exposure, the stopper was removed and mice allowed to recover. The time under anesthesia was about 90 min. Comparison of the distribution of turmerones in mice was done as follows: Amount of each turmerone in each organ/summation of that turmerone in all organs.

2.5. Sample collection

The blood (1 mL) was obtained from the inferior vena cava while anesthetized with the sodium pentobarbital. The left cardiac ventricle of the mice was subsequently perfused with an isotonic sodium chloride solution. The lung, olfactory bulb, brain (cerebrum and cerebellum), heart, liver, kidney, epididymal fat, and brown adipose tissue were isolated and subsequently frozen on a plate cooled with liquid nitrogen without any chemical fixation. These organ samples were stored at -80 °C for a maximum of 2 wk.

2.6. LC-MS/MS analysis

Turmerones in serum or organs were measured using an ultra-performance liquid chromatography-mass spectrometry (UPLC-MS) system (LC-MS-8050; Shimadzu Co., Ltd., equipped with Lab Solutions ver. 5.75). For turmerone analyses, the experimental conditions were as follows: two columns (L-Column2 ODS; length: 150 mm, particle size: 2.0 µm, I.D.: 2.1 mm, Chemicals Evaluation and Research Institute) in series were used; flow rate: 0.4 mL/min, mobile phase: water:acetonitrile (18:82), retention time: 7.5 min for α -turmerone, 7.3 min for β -turmerone and 5.8 min for ar-turmerone, and injection volume: 1 µL. The electrospray ionization mass spectrometry (ESI-MS) parameters were as follows: interface voltage: 4.0 kV, ionization mode: positive mode, source temperature: 400 °C, desolvation gas flow: 10 L/min and desolvation temperature: 100 °C. Argon was used as a collision-induced dissociation gas at 270 kPa, with a collision energy of 14 eV. The multiple reaction monitoring (MRM) mode was used to monitor the transition of $[M+H]^+ = m/z$ 219.10 to $[C_9H_{13}]^+ = m/z$ 121.10 with α -turmerone, $[M+H]^+ = m/z$ 219.10 to $[C_9H_{13}]^+ = m/z$ 121.10 with

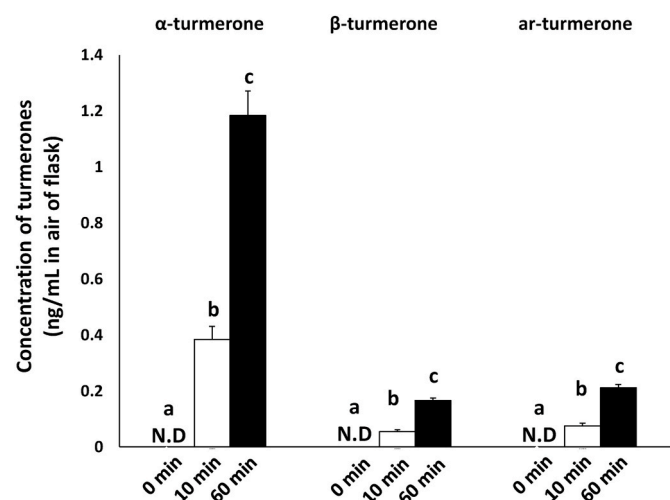


Fig. 1. Concentration of volatile turmerones (α-turmerone, β-turmerone, and ar-turmerone) in a flask. Control group (n = 6), turmeric oil group (n = 7). Values with different letters are significantly different (p < 0.05).

β-turmerone, and that of $[M+H]^+ = m/z$ 217.35 to $[C_9H_{11}]^+ = m/z$ 119.20 with ar-turmerone.

2.7. Statistical analyses

The values were expressed as the mean ± standard deviation (S.D.). Statistical differences were determined using one-way ANOVA with the Scheffe test or Dunnett test. A p < 0.05 was considered statistically significant. Statistical analyses were done using the Stat View 5.0 software (SAS Institute, Cary, NC, USA).

3. Results

3.1. Volatile turmerone concentrations in the flask

The concentrations of volatile turmerones in a flask without mice was investigated (Fig. 1). Turmerones were not found at 0 min. After 10 and 60 min, α-turmerone, β-turmerone, and ar-turmerone increased in a time-dependent manner. Among the 3 turmerone species, α-turmerone was dominant.

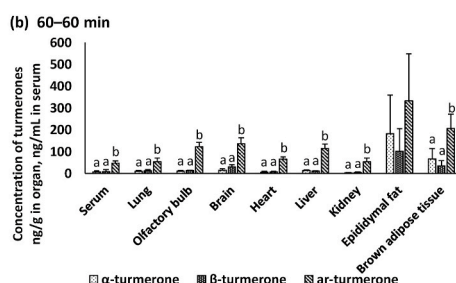
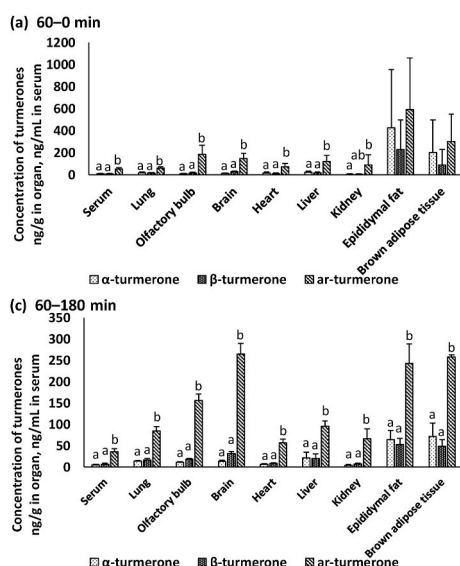


Fig. 2. Turmerones concentrations in mice. Before exposure, flask was closed with a silicone stopper for 10 min to fill it with volatile turmerones. Mice were exposed to volatile turmerones for 60 min. After 60 min of exposure while anesthetized, the stopper was removed and mice allowed to recover. The experimental scheme and conditions are shown in Supplementary Fig. 1. (a) 60-0 min group (n = 6), (b) 60-60 min group (n = 6), (c) 60-180 min group (n = 6). To adjust the background level, averaged values of the negative control group were subtracted from values for the volatile turmeric oil exposure groups. Values with different letters are significantly different (p < 0.05).

3.2. Serum and organ turmerones concentrations in mice

The distribution of turmerones in the mice with several experimental conditions were estimated. The distribution of turmerones in the mice exposed to volatile turmerones was quantified with 3 different conditions, i.e., the 60-0, 60-60, and 60-180 min groups. All turmerones were found in all the organ samples (Fig. 2). The distribution of ar-turmerone was significantly higher or had a higher trend in all the organ samples than that of α-turmerone and β-turmerone. There were no significant differences between the distribution of α-turmerone or β-turmerone in all the organ samples.

3.3. Temporal change in turmerone concentrations in mice

To estimate residual turmerones in the mice, the turmerone amounts in the mice was quantified until 72 h after exposure. All turmerones were found in all the samples until 180 min after exposure (Fig. 3 (a)-(i)). In all the organ samples, α-turmerone and β-turmerone were not found 24 h after exposure. Whereas ar-turmerone was found in the serum, lung, olfactory bulb, brain, liver, kidney, and epididymal fat but not in the heart and brown adipose tissue. No turmerones were found in the organ samples 72 h after exposure (Fig. 3 (j)).

3.4. Intra-organ comparison of the distributed turmerone compositions

The turmerone compositions were investigated to estimate the organ distribution properties of each turmerone. Due to high individual mouse differences, the adipose tissues were excluded in this comparison. In the 60-0, 60-60, and 60-180 min groups, the α-turmerone in the kidney was significantly lower or had a lower trend than that of the other organs (Fig. 4 (a)-(c)). In the 60-60 min group, the α-turmerone in the heart was significantly lower than that in the lung, olfactory bulb, brain, and liver (Fig. 4 (b)). In the 60-0 min group, the β-turmerone in the brain was significantly higher than that in the other organs (Fig. 4 (d)). In the 60-60 and 60-180 min groups, the β-turmerone in the brain was significantly higher or had a higher trend than that in the other organs (Fig. 4 (e), (f)). In the 60-60 and 60-180 min groups, the β-turmerone in the heart and kidney were significantly lower or had a lower trend than those in the other organs (Fig. 4 (e), (f)). In the 60-0, 60-60 and 60-180 min, and the 60-24 h groups, the ar-turmerone in the olfactory bulb was significantly higher or had a higher trend than that in the other organs (Fig. 4 (g)-(j)). In the 60-60 min group, the ar-turmerone in the liver was significantly higher than that in the lung, heart, and kidney (Fig. 4

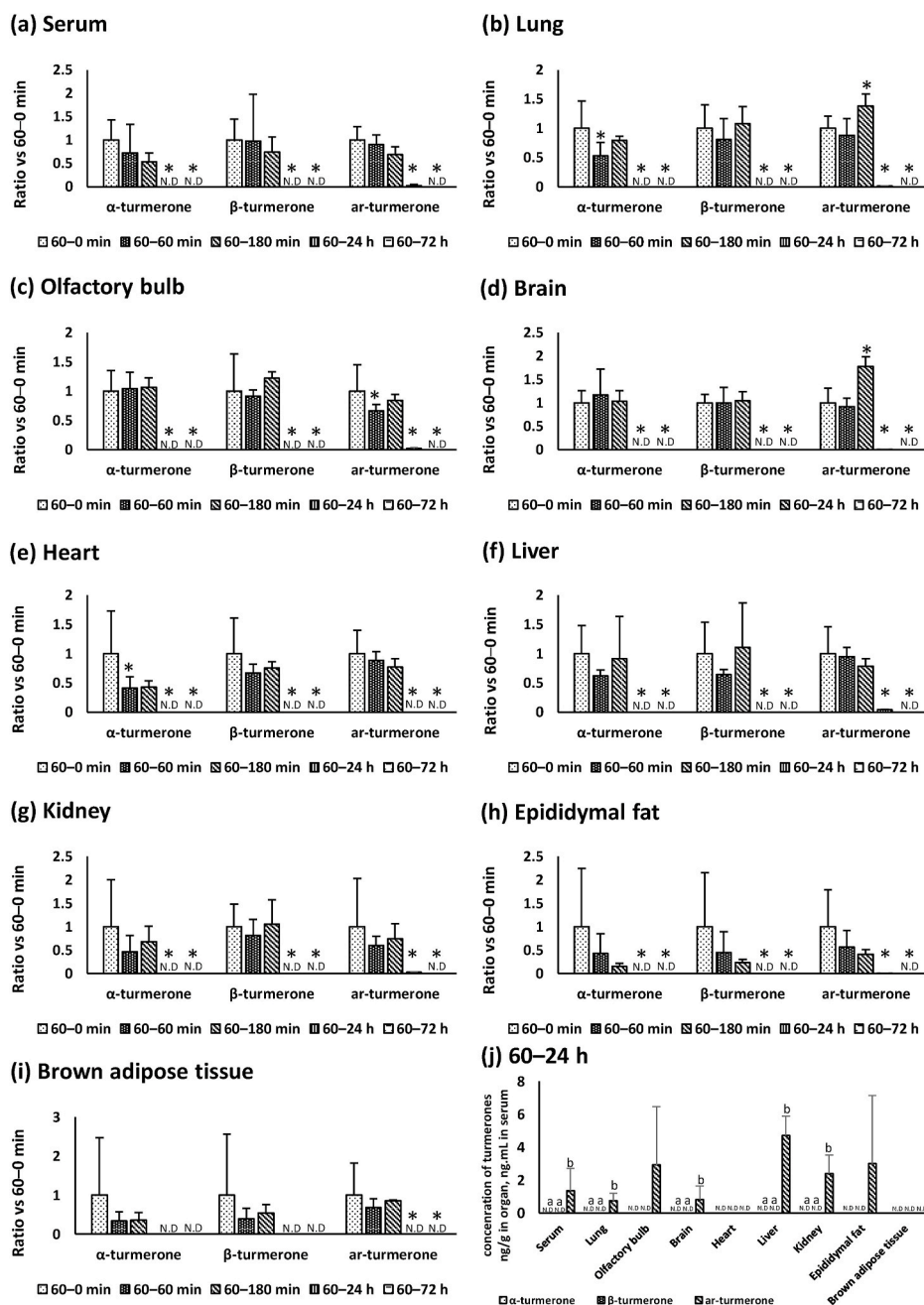


Fig. 3. Temporal changes of turmerones in mice. (a) Serum, (b) Lung, (c) Olfactory bulb, (d) Brain, (e) Heart, (f) Liver, (g) Kidney, (h) Epididymal fat, (i) Brown adipose tissue, (j) Turmerones concentrations in mice in 60-24 h group. 60-0 min (n = 6), 60-60 min (n = 6), 60-180 min (n = 6), 60-24 h (n = 6), and 60-72 h group (n = 7). Values are shown as the ratio versus the 60-0 min group. Values of turmerones in each 60-0 min group were set as 1. To adjust the background level, averaged values of the negative control group were subtracted from values for the volatile turmeric oil exposure groups. Values with different letters are significantly different ($p < 0.05$). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

(h)). In the 60-180 min group, the ar-turmerone in the brain was significantly higher than that in the other organs (Fig. 4 (i)).

3.5. Intra-turmerone comparison of the composition in the organs of mice

The turmerone compositions in the mouse organs were compared between the 3 turmerones to estimate the difference in the distribution properties of the organs. In the 60-0 min group, the α -turmerone in the lung was significantly higher than the ar-turmerone (Fig. 5 (a)). The ar-turmerone in the olfactory bulb was significantly higher than the α -turmerone in the 60-0 and 60-180 min groups and the β -turmerone in the 60-0 min group (Fig. 5 (a)). In the 60-0 and 60-60 min groups, the β -turmerone in the brain was significantly higher than the α -turmerone and ar-turmerone (Fig. 5 (a), (b)). In the 60-180 min group, the ar-turmerone in the brain was significantly higher than the α -turmerone (Fig. 5 (c)). In the 60-24 h group, the ar-turmerone was significantly

higher than the α -turmerone and β -turmerone in the lung, olfactory bulb, brain, liver, and kidney (Fig. 5 (d)).

4. Discussions

Turmerones are components of essential oils derived from turmeric rhizomes, and recent studies have shown their various biological activities as functional food compounds. This study focused on turmerones as volatile compounds and estimated their distribution in mice exposed to volatile turmerones. With experimental conditions, intact turmerones were found in all the mice's organs. Turmerones have characteristic distribution properties, suggesting that different chemical structures affect the distribution in organs and the residual time.

Among the 3 turmerone molecules, α -turmerone was dominant in the flask without mice, whereas ar-turmerone was the most abundant in the mice. The α -turmerone level in the mice was nearly equal to the

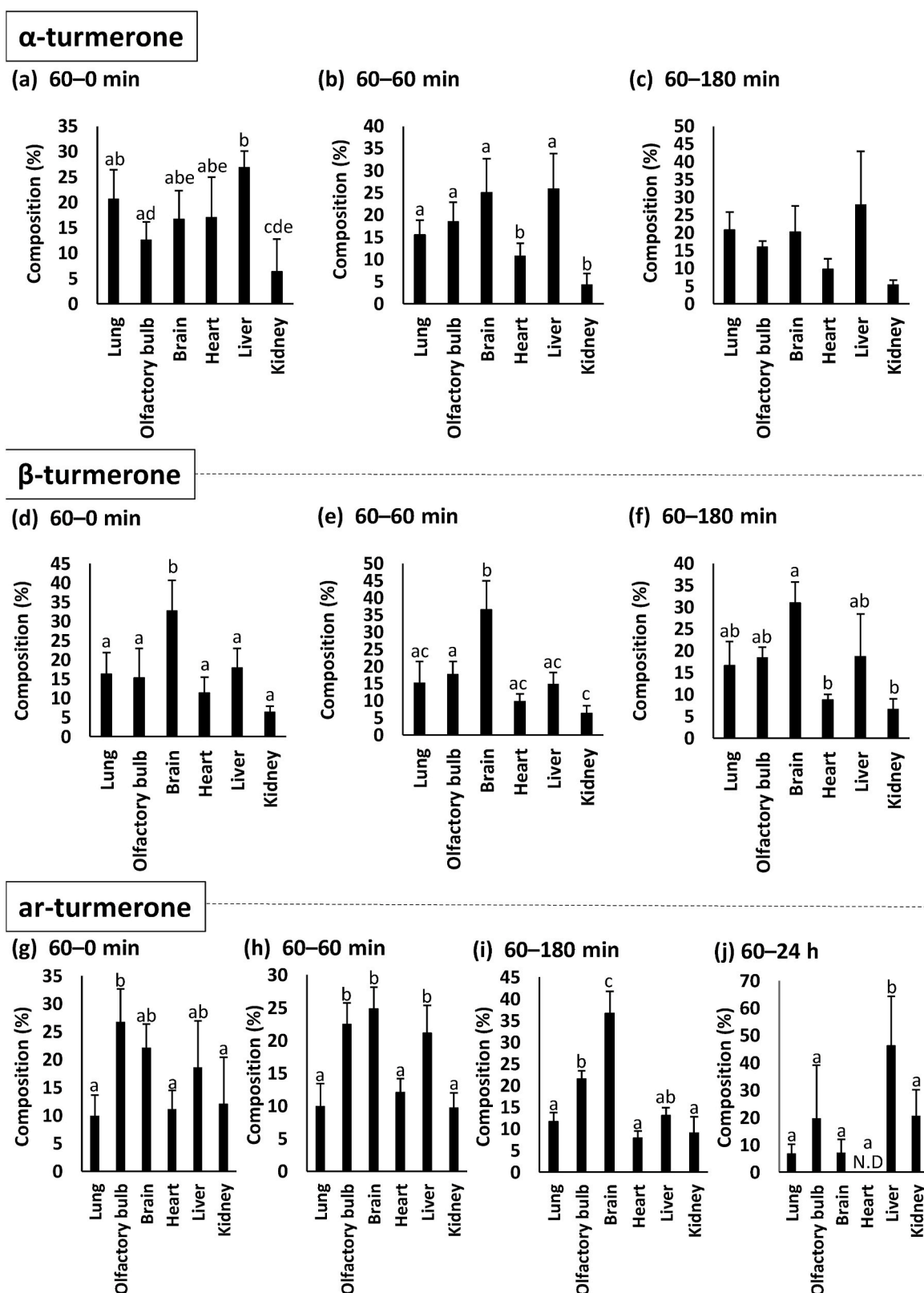


Fig. 4. Comparison of the distribution of turmerones in mice. Comparison of the distribution of α -turmerone in mice. (a) 60–0 min group ($n = 6$), (b) 60–60 min group ($n = 6$), and (c) 60–180 min group ($n = 6$). Comparison of the distribution of β -turmerone in mice. (d) 60–0 min group ($n = 6$), (e) 60–60 min group ($n = 6$), and (f) 60–180 min group ($n = 6$). Comparison of the distribution of ar-turmerone in mice. (g) 60–0 min group ($n = 6$), (h) 60–60 min group ($n = 6$), (i) 60–180 min group ($n = 6$), and (j) 60–24 h ($n = 6$). To adjust the background level, averaged values of the negative control group were subtracted from values for the volatile turmeric oil exposure groups. Composition values are calculated using the equation. Amount of each turmerone in each organ/summation of that turmerone in all organs. Values with different letters are significantly different ($p < 0.05$).

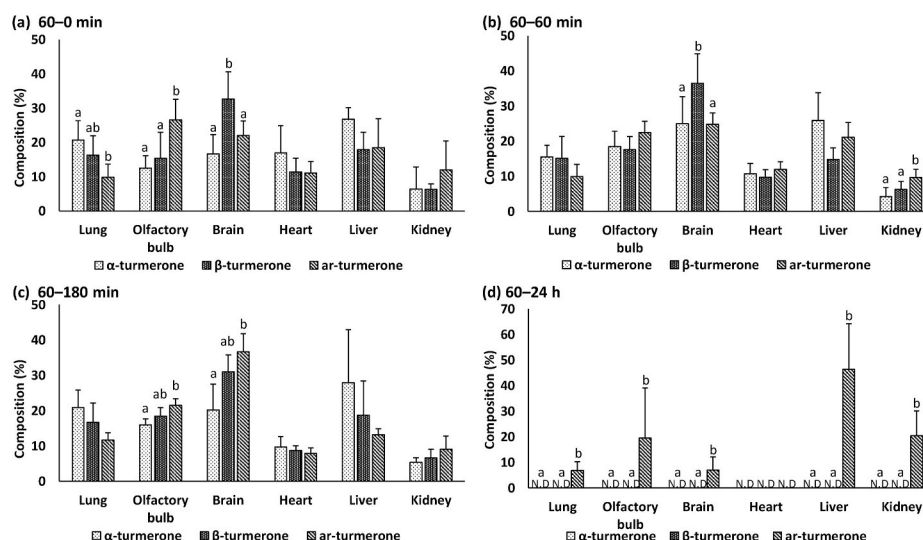


Fig. 5. Comparison of composition among turmerones. (a) 60–0 min group (n = 6), (b) 60–60 min group (n = 6), (c) 60–180 min group (n = 6), and (d) 60–24 h (n = 6). To adjust the background level, averaged values of the negative control group were subtracted from values for the volatile turmeric oil exposure groups. Values with different letters are significantly different (p < 0.05).

β-turmerone level. From a structural point of view, ar-turmerone with a benzene ring is more stable than α-turmerone and β-turmerone. The conversion reaction of α-turmerone and/or β-turmerone to ar-turmerone requires an oxidizing agent or a metal catalyst. The predominant amount of α-turmerone in the flask suggested that conversion from α-turmerone and/or β-turmerone to ar-turmerone does not proceed at the animal's room temperature. Unidentified enzymes may contribute to the conversion of α-turmerone and/or β-turmerone to ar-turmerone in the mice. The enzyme might be ubiquitous in mice because ar-turmerone was observed at relatively high concentration in all the organs.

In previous report, α-turmerone and β-turmerone were found at concentrations of 2 and 1 ng/mL in serum of 5 wk old male ddY mice which were orally administered turmeric oil at 22.5 μg/g (bw) (Iwamoto et al., 2019). Results of this study suggested that the distribution between oral administration and inhalation was not different.

Volatile turmerones might be incorporated into the mice's organs from the microvessel in the lung. In the brain, the blood-brain barrier limits the transfer of compounds from the blood to the brain. In general, only a few functional food compounds can be distributed to the brain. The distribution of orally administered turmerones in the brain, suggested that turmerones cross the blood-brain barrier (Iwamoto et al., 2019; Matsumura et al., 2016). Inhaled volatile turmerones may be distributed into the brain from the blood circulation. Kumar et al. (1982) showed the possibility of the distribution of nasally administered pharmaceuticals to both the cerebrospinal fluid and the blood. Direct distribution by the nasal cavity might be a potential pathway to the brain. However, the mechanisms underlying the distribution of volatile turmerone remain unknown. Further studies on these mechanisms are needed.

The turmerone concentrations were much higher in the epididymal fat than in the other organs. This may be due to the lipophilic nature of turmerones. The turmeric extract reportedly stimulates human adipocyte differentiation *in vitro*, and ar-turmerone has a ligand-binding activity for peroxisome proliferator-activated receptor-γ (Kuroda et al., 2005). Inhaled turmerones may affect the adipose tissue function *in vivo* as well as orally administered because the turmeric ethanol extract suppressed the blood glucose elevation in type 2 diabetic KK-Ay mice (Kuroda et al., 2005). In addition, as explained in the introduction, turmerones have multiple biological activities so it is important to evaluate the effects of the inhaled turmerones on their physiology.

5. Conclusions

Inhaled bioactive turmerones were distributed in their intact forms in animal bodies, and ar-turmerone residuals were found for 24 h. In environments exposed to high concentrations of volatile components, such as food factories, it is necessary to assess the effects on living organisms. A limitation of this study was the lack of an effort to determine mechanisms underlying the distribution of volatile turmerones in organs. Further studies on these mechanisms are needed.

Authors' contributions

Y.T. and T.S. did animal experiments, measured the data and wrote the manuscript. C.K. did the animal experiments and edited the manuscript. S.M. did quantitative analysis of turmerones. Y.Y. and S.M. obtained the LC-MS data and edited the manuscript. T.M. edited the manuscript. N.Z. designed the studies, wrote and edited the manuscript, and secured the funding.

Declaration of competing interest

The authors confirm that they have no conflicts of interest with respect to the work described in this manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fbio.2021.100965>.

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